

An Effect of Limb Position in Motor Imagery Training Paradigm in Immersive Virtual Environment

Suktipol Kiatthaveephong, Suvichak Santiwongkarn¹, Rattanaphon Chaisaen, Chutimon Rungsilp
Tohru Yagi² and Theerawit Wilaiprasitporn

Vidyasirimedhi Institute of Science and Technology (VISTEC), Thailand

¹Sirindhorn International Institute of Technology (SIIT), Thammasat University, Thailand

²Tokyo Institute of Technology, Tokyo, Japan

Abstract—Motor imagery (MI) training-based Brain-Computer Interfaces (BCI) improved individuals' motor function by inducing direct alterations in the sensorimotor area. Virtual Reality (VR)-based MI training has been identified as a promising technique for achieving high performance. However, the physical limb position should be considered when designing a better training task. This paper investigated the effect of induced MI activities when the virtual arms were at normal and shifted down position. The paradigm used of Virtual Reality (VR) to simulate the situation of having realistic and unrealistic arms position in an immersive virtual environment. Analyses of electroencephalograms (EEGs) revealed significant differences in MI activity levels between two positions on both the left and right sides. During shifted arms MI, the negative power regions were found in beta and gamma bands on the contralateral hemisphere in time-frequency analysis. Resting vs. normal position arms MI and resting vs. shifted position arms MI classification accuracy reached 80% and 63%, respectively. Overall, these findings suggested that taking into account the realistic physical position of virtual limbs is critical for optimizing MI training performance.

Key Words—Brain-Computer Interfaces (BCI), electroencephalogram (EEG), motor imagery (MI), virtual reality (VR)

I. INTRODUCTION

An effective rehabilitation method for strokes is a closed-loop brain-computer interface (BCI) based on motor imagery (MI). It has been utilized as a training method to restore and improve motor function in patients and healthy individuals through direct communication between the human brain and external devices [1]. The training involves users repeatedly imagining the movement of real body parts to trigger cortical modulation at the sensorimotor cortex. However, standard MI training based on a monitor screen may not produce the best results. Virtual Reality (VR) technology has been hailed to be the promising method for MI training [2] because of its immersiveness and countless approach to construct training tasks for generating MI-EEG. The previous research [3] suggested that using an immersive VR headset induced MI activities rather than a simple monitor screen, giving a better sense of embodiment during the MI training. A study suggested that the action observation-based rehabilitation should concern the similarity of physical appearance for effective MI training paradigm [4]. However, virtual reality was not fully compatible with the actual environment. For example, the misperceived scale and distance in VR [5]. Nonetheless, there was no evidence about

the imagery of the shifted arm from the actual positions in virtual reality.

In this work, we proposed another possible factor that affects motor imagery training using the combination of Electroencephalogram (EEG) [6] and VR technology. We simulated the virtual arms at normal and shifted positions to observe the effects on MI activities.

II. METHOD

A. Participants

Ten right-handed subjects (mean age $\pm$ std=27.7 $\pm$ 7.67; one female) were recruited for an experiment. They were naive to MI-BCI. Six of them had experience with Head-Mounted Display-VR (HMD-VR). All subjects provided written informed consent in accordance with the Declaration of Helsinki. Then, the experimenter gave instructions to all subjects before the experiment began.

B. Data acquisition

EEG signals were amplified and recorded with a sampling frequency of 1200 Hz using an amplifier (g.Hiamp, g.tec, Austria). Sixty-two electrodes were positioned following the 10-20 international system. FP_z and right ear clips were used as ground and reference electrodes, respectively. The impedance of all electrodes was kept below 30 k Ω before putting on HMD-VR (Oculus Quest 2, Oculus, USA). Oculus Quest 2 showed the immersive virtual environment to the subjects throughout the experiment.

In the virtual scene, the subject transformed into a human avatar with four virtual arms. Two of which were in the normal position (NP) and the other two were extended from its body below the NP arms, called shifted position (SP) arms. The appearance and dimension of the virtual arms were identical. The scene's background was black to avoid distraction. The subjects could only see the avatar's body and visual cues.

C. Experimental Protocol

The experimental protocol was divided into four sessions, as illustrated in Figure 1. Each session consisted of 50 trials that last 10 minutes. In each trial, the subjects had to complete the tasks based on the visual cue. The subjects were then advised to rest at least three minutes after each session. When the trial began, the subjects had to look at the white fixation cross

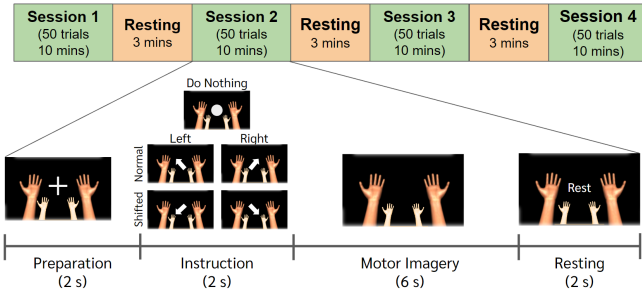


Fig. 1. An experimental protocol

for two seconds (preparation). Following that, one of the five visual cues was shown randomly for two seconds to inform the subject of what they needed to do next (Instruction). Four of the five visual cues were different diagonal arrows. The up-left, up-right, down-left, and down-right arrows represented left-NP (L-NP), right-NP (R-NP), left-SP (L-SP), and right-SP (R-SP) movements imagination, respectively. Furthermore, the white circle indicated that the subjects could be relaxed and do nothing for that trial. The visual cue then disappeared after six seconds, and the subjects performed Kinesthetic Motor Imagery (KMI) based on the visual cue that they had previously received (Motor Imagery). Finally, 2-second “Resting” word displayed to allow the subjects to relax and prepare for the next trial (Resting).

D. EEG Pre-processing

The raw EEG signals were band-pass filtered from 1 to 40 Hz. Then we downsampled the filtered signal to 250 Hz. Independent component analysis (ICA) was also applied. Bad channels were identified, removed, and interpolated using pyprep [7]. Subsequently, we extracted the epochs from -4 to 6 at the trigger onset and then obtained all 200 trials per subject, which contained 40 trials per single task according to an experimental protocol.

E. EEG Analysis

1) *ERD/ERS*: The following equation was used to calculate the levels of Event-Related Desynchronization (ERD) and Event-Related Synchronization (ERS) [8] to determine how much the subjects induced MI activities for each task.

$$ERD/ERS(f) = \frac{PSD_{MI}(f, t_m) - PSD_{base}(f, t_b)}{PSD_{base}(f, t_b)} \quad (1)$$

where $PSD_{MI}(f, t_m)$ is the averaged Power Spectral Density (PSD) of the specific frequency f during the motor imagery task, t_m , between 0 and 6 seconds from the trigger onset. The frequency power during the baseline period $PSD_{base}(f, t_b)$ was also computed by averaging power within the baseline interval, t_b , between -4 and -2 from the trigger onset. The multitaper approach was used to calculate PSD on three frequency bands, which were mu (8-13 Hz), beta (13-30 Hz), and gamma (30-40 Hz).

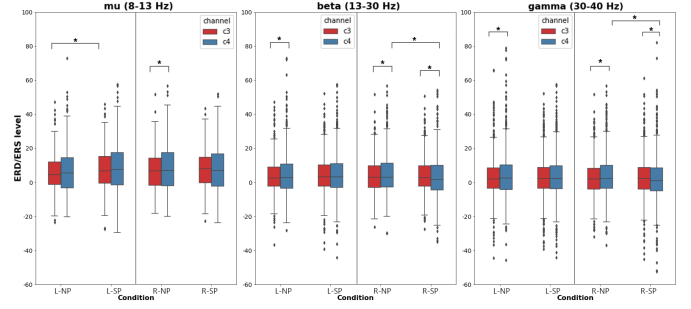


Fig. 2. The ERD/ERS levels box plots of C3 (red) and C4 (blue) channels among four MI tasks on the specific frequency bands, including mu (8-13 Hz), beta (13-30 Hz), and gamma (30-40 Hz). The significant differences were labeled as an asterisk (*) if $p < 0.05$

The statistical tests for three frequency bands were performed on the pairs (mu, beta, and gamma). The Wilcoxon rank-sum test was used to see if there were any significant differences in the ERD/ERS levels of four different pairs (pairs between NP and SP arms on the same lateral for each channel). The Wilcoxon signed-rank tests were then applied to the remaining four pairs (pairs of C3 and C4 channels for each MI task). As a result, a total of eight pairs were completed.

2) *Time-Frequency Analysis*: The time-frequency analysis focused on the dynamic modulations of the oscillatory activities before and after the onset of events. Morlet wavelets were used to decompose this spectro-temporal information across the interval of interests. The decomposition was then performed before correcting baseline power from -4 to -2 seconds from trigger onset.

F. Motor Imagery Classification

The classification was done in a subject-dependent manner which consisted of two parts:

First, filter bank common spatial pattern (FBCSP) was used to extract EEG signal features by using a spatiotemporal discriminator based on mutual information of multiple frequency bands [9]. The pre-processed EEG signal was decomposed into nine frequency bands between 3 and 40 Hz. The decomposed signal was input to the common spatial pattern (CSP) algorithm. The feature vector resulted from the common spatial patterns by using discriminator mutual information of the feature.

Second, support vector machine (SVM) was used to classify the feature vectors. This step focused on classifying resting state and lateral motor imagery in three aspects:

- Rest, L-NP MI, and R-NP MI
- Rest, L-SP MI, and R-SP MI
- Rest, Left-arm MI, and Right-arm MI

III. RESULTS AND DISCUSSION

A. ERD/ERS results

To evaluate the MI activities induced during the different MI tasks, the ERD/ERS levels were used as the quantitative measurement of how much ERD and ERS were caused compared to the baseline.

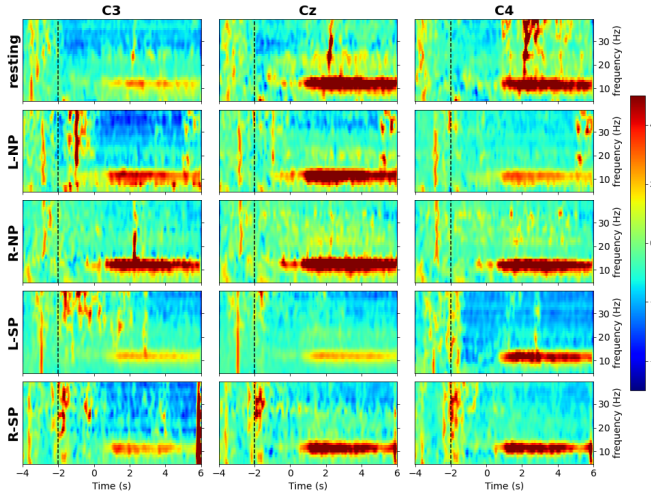


Fig. 3. The grand average time-frequency plots of C3, Cz, and C4 channels among all tasks. The baseline powers were computed by using the time interval before the vertical dashed line.

The box plots in Figure 2 depicted the levels of ERD and ERS on mu, beta, and gamma bands for C3 and C4 channels. The pairs with statistical significance would be labeled with an asterisk (*) if those p values were less than 0.05. ERD/ERS levels were lower in the beta and gamma bands compared to the mu bands. The statistical Wilcoxon test revealed significant differences between channel C3 and C4 on the L-NP task (beta and gamma bands: $p < 0.05$), R-NP task (mu: $p < 0.01$; beta: $p < 0.001$; gamma: $p < 0.0001$), and R-SP task (beta: $p < 0.05$; gamma: $p < 0.01$). The differences between the two MI tasks were statistically significant on mu (left MI tasks; C3 channel; $p < 0.05$), beta (right MI tasks; C4 channel; $p < 0.05$), and gamma (right MI tasks; C4 channel; $p < 0.01$).

The statistical analyses showed that the NP and SP motor imagery from the same lateral has significant differences. This finding indicated that the effects of ERD/ERS levels of imagining arm movement in a normal position differed from an unrealistic position.

B. Time-Frequency Analysis

The grand average time-frequency plots of all subjects are illustrated in Figure 3. Each plot represented the oscillatory changes in the C3, Cz, and C4 channels for a single task. The increased mu power was discovered after the trigger was initiated. Meanwhile, mu power was slightly decreased on the opposite hemisphere to the limb performing the MI task. Furthermore, negative power covered the beta and gamma bands on the opposite side, which was most noticeable on the L-SP and R-SP tasks.

The presence of high positive mu power was unclear from the time-frequency results. Previous studies discovered similar activities, obviously on the visual cortex, during visual imagery [10] and MI task compared to MI + action observation task [4]. However, the reduction in power in the mu, beta, and

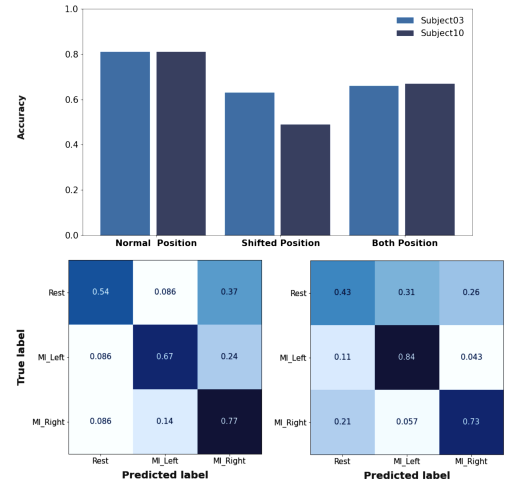


Fig. 4. Classification result of lateral motor imagery tasks of two subjects from FBCSP+SVM classifier with 7-Fold CV (Top) and their confusion matrices from lateral motor imagery classification (Bottom-left for subject03 and Bottom-right for subject10)

gamma bands on the contralateral rather than the ipsilateral hemisphere corresponded to previous MI research [3], [11].

C. Classification Results

From the 3-class classification by FBCSP+SVM, the average accuracy of ten subjects was 0.42 ± 0.11 . The lowest accuracy was 0.27, and the highest accuracy was 0.81. There were two subjects with classification accuracy above 0.80. This inferred that the subjects' imagery consists of distinctive features between rest state, L-NP imagery, and R-NP imagery. For shifted arm imagery, both subjects' classification resulted in lower accuracy than NP MI. Therefore, the shifted arm MI consists of lateral features in EEG signal in the same manner but lower level than normal arm MI, as shown in Figure 4.

D. Limitation

There were no quantitative or qualitative measures in place during the MI tasks to ensure that each trial was not impacted by muscle contraction. Electromyography (EMG) [12] should be used to validate the pure MI trial using a specific EMG activity threshold before analyzing EEG.

IV. CONCLUSION

In this paper, we proposed the effect of the virtual arm setup position on motor imagery. Virtual reality was used to simulate virtual arms at normal and shifted positions to observe the MI activities. The ERD/ERS and time-frequency results revealed the dissimilar pattern of EEG signals between the normal arms MI and shifted arms MI. Only the negative power of the beta and gamma bands could distinguish between normal and shifted limbs. The classification results confirmed the discrimination level of shifted limbs was lower than the normal limb positions. Overall results show that the position of arms in VR effects the MI quality. Eventually, the position of virtual limbs should be considered the critical factor in designing an effective MI training task in the virtual environment.

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